Introduction to Computer Science

Diploma in Software Engineering 24.3

Coursework - Software Engineering



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❖ Desktop Computer Specification

- **Casing (Tower)**
  - The casing is the external case that can see in Computer.
  - There four types of cases,
    1. Full Tower
    2. Mid Tower
    3. Mini Tower
    4. Micro Tower
  - It's an important part of the system, as it provides physical protection for the hardware, part inside it.
  - Casing help with airflow and to reduce the overheating.
  - The tower case servers mainly as a way to physically mount and contain all the actual components.

- **Processor (i7 11th Generation)**
  - The processor (CPU) is the brain of the computer, responsible for executing instruction.
  - The intel core i7 -11700 is an 11th generation processor, it is a high – performance CPU aimed at gamers, content creators and power users.
  - Intel core i7 -11700 11th generation has 8 cores; the number of cores allows the CPU to handle multiple tasks at ones.

- And 16 treads supports hyper threading and also has 16MB cach.This is a small , fast memory built, used to store frequently accessed data for quicker retrieval.

- **Chipset (intel Q570)**
  - The Chipset is a crucial part of the motherboard, that controls communication between the CPU, RAM, storage and other components.
  - It's supports intel's 10th and 11th generation core processor, such as the core i5, i7, i9 series.
  - The motherboard supporting both DDR4 and LPDDR4, and up to 64GB of memory, Also the intel Q570 chipset is flexible in terms of RAM.

- **Operating System**
  - An operating system is software that manage computer hardware and software resources and provides common services for computer programs.
  - And windows 11 pro is the latest version on Microsoft's operating system. Enhanced features like better performance Multitasking capacities than other operating system**.**

- **RAM**
  - RAM is temporary storage that allows the computer to quickly access data for running application.

- 16GB of capacity is a good balance for most tasks, content creation, office work and better gaming.
- Also, DDR4 Technology is the 4th generation of DDR memory, improved performance over its predecessor.
- 3200MHz is the type and speed of the RAM 3200MHz refers to the speed at which it can transfer data.

• DIMM

- DIMM slots refers to the number of **Dual Inline Memory Module (DIMM)**
- 4 slots the motherboard can support up to four separate memory sticker, and there are deferent types of DIMM slots for DDR3, DDR4 etc.
- Having 4 DIMM slots provides flexibility for upgrading memory or expanding system capacity over time.

• SSD

- A 512GB SSD is a solid-state Drive capacity of keeping 512 Gigabytes of data.
- Solid state Drives are faster than **HDDS (Hard Disk Drives)** so loading and booting times are reduced on them.
- SSD are better for Gaming and heavy suites, but generally cost a lot more.

- ## HDD

  - 1TB HDD with 7200RPM SATA refers to a 1TB **Hard Disk Drive (HDD)** than spins at 7200 **Revolutions Per Minute (RPM)** and **SATA (Serial ATA)** interface for data transfer.
  - A 1TB HDD with 7200RPM offers a good balance between storage capacity, performance, and price.
  - The general – purpose storage needs, though not as fast as an SSD.

- ## WLAN

  - WLAN 801.11ac is a WIFI used for wireless networking, offering faster speeds and improved performance compared earlier standard like 802.11n.
  - It's provides fast, reliable wireless internet access, especially beneficial for high bandwidth activities.

- ## USB Version 3

  - USB known as Super Speed USB.
  - USB 3.0 marked a significant leap in data transfer speed and power delivery, making it suitable for modern high-speed storage devices.
  - USB port is used to connect smartphones, computers and other electrical devices.

- ## USB TYPE C

  - USB TYPE C is flexible and widely adopted connector a variety of devices.
  - Here are some common users for USB TYPE C,

    1. Charging Devices
    2. Data Transfer
    3. Video
    4. Audio Devices
    5. Networking

  - USB TYPE C's universal and flexible design allows it to replace a variety of older connectors, data and video functions into single port.

- ## HDMI

  - **HDMI (High- Definition Multimedia Interface)** is commonly used to connect devices like TVs, Monitor, Laptops, Blu- ray players.
  - HDMI supports high definition (HD), and even 8K video resolutions, as well as high-quality audio format.
  - Also, generally used standard for transmitting high-quality audio and video signals between devices.

- ## External Port (RJ45 PORT)

  - The RJ45 port is type of connector commonly used for Ethernet Networking.

- It typically connects devices like computer, routers, switches and other network hardware to a **Local Area Network (LAN).**
- Also, an RJ45 port with 10/100/1000 Mbps support allows you to connect to a wired network, supporting varying speeds depending on the network devices connects.

- **Pointing Device**
  - An optical mouse with scroll button is a type of computer mouse that uses optical sensor technology to detect movement.
  - The optical mouse uses an LED light and sensor to detect movement across a surface.

- **Key Board**
  - In the US English layout, the primary differences are placement of punctuation marks, symbols, and special characters specific to the language.
  - The key board includes all these elements, as opposed to more compact layout like smaller keyboards without the number keyboard.

- **Display**
  - A 23.8” LED display offers a 23.8-inch diagonal screen size with LED backlighting, delivering sharp vibrant visual while being energy-efficiency

- It is ideal for general comporting, office work and casual entertainment, providing a good balance between screen size and space efficiency.

- **Warranty**
  - A 3 years' warranty (Part and Labor) means that for 3 years, the manufacture will cover both the cost of replacing faulty parts and the labor required to repair or replace the product.
  - This provides peace of mind, ensuring that any defects or issue that arise during the warranty period will be handled at no extra cost to the customer.

❖ Best Hardware Setups for Fast Desktops, Laptops, and Servers use by a graphic agency

Hardware specification for desktops

DESKTOPS

- **Processor (intel i7 10th generation)**
  - The Intel core i7 10th generation is a high- performance processor with 6 to 8 cores, Hyper- Threading and turbo boost up to 5.0GHz, offering strong multitasking and gaming capabilities.

- It can run video editing and graphic design smoothly.
- These processors will allow the design team to run software efficiently without delays.

- **RAM (32GB- DDRM)**
  - Graphic design work requires a large amount of RAM 32GB DDR4 RAM provides faster data processing and handling, demanding, applications like video editing, gaming and virtual machines.
  - 32GB of RAM will help the designers avoid system slowdowns and work smoothly with resource-heavy files.

- **Storage (1TB SSD)**
  - 1TB SSD provides fast data access, significantly improving boot times, file transfer speeds, and overall system performance compared to traditional hard drives.

- **Graphic Card**
  - The NVIDA GeForce RTX 3060 or better is a high-performance graphics card that significantly boost the visual and computational power.
  - It helps with video editing, 3D rendering, and gaming by making graphics load faster and more smoothing.
  - It’s great choice for anyone who needs extra power for creative or technic.

Hardware specification for laptops

LAPTOPS

- **Processor (intel i7 10th generation)**
  - This is powerful processors that can handle demanding task video editing, gaming and multitasking without slowing down.
  - Perfect for users who need both speed and efficiency for everyday work or creative projects.

- **RAM (16GB DDR4)**
  - This RAM is great amount of memory for everyday task it allows your computer to quickly access data and smoothly switch between applications without slowing down.
  - Its sticker a good balance between performance and cost for most users.

- **Display (15.6-inch full HD 1PS display)**
  - Full HD means images and text appear crisp, making it great for work, watching videos, or editing photos.
  - So, whether you're working on design projects or streaming content, the picture stays consistent and accurate.

- **Storage (512GB SSD)**
  - 512GB SSD provides fast data access, significantly improving boot times, file transfer speeds, and overall system performance compared to traditional hard drives.

Hardware specification for servers

Server

- **Processor (Intel Xeon Silver or AMD**
  - This processor with 8 scores or more is designed for servers and heavy workloads.
  - These processors can handle multiple task at once, making them perfect for running websites managing database or hasting applications.
  - These are ideal choices for business that need powerful, always- on system for demanding task.

- ## Network Inference (10G Network Adapter)
    - This Network adapter allows your server to transfer data much faster than standard network connection, reaching speeds of 10GB per second.
    - It supports smooth, high-speed connections even when multiple users or system are accessing the network at the same time.

- ## RAM (64GB DDR4 ECC)
    - **64GB DDR4 ECC RAM** is high-capacity memory with built in error correction, making it ideal for servers or systems that handle critical data.
    - **ECC RAM** automatically detects and corrects memory errors, preventing data corruption and system crashes.
    - **64GB,** your system can handle large amounts of data, run multiple applications and making it perfect for businesses that need reliable, always - on performance.

- ## Storage (2TB SSD)
    - This storage provides fast storage with redundancy, meaning your data is mirrored or distributed across multiple drives to prevent loss if one drive fail.
    - This setup ensures quick access to data, high reliability, and protection against hardware failures, keeping your system running smoothly and your data safe.

❖ Operating system and application software used by graphic agency

Operating System

- **MacOS:** Many designers prefer macOS because it offers stability, Great color accuracy, and design work like photo editing and illustration

- **Windows:** Some agencies use Windows because it offers flexibility in Hardware and can be more cost effective. It supports all major design software, and some designers prefer. The wider range of custom- built PCs available.

Application System

- **Core IDRAW:** A popular alternative to illustrator for vector- Base design**,** especially in print and logo creation.
- **Sketch:** Used primarily for web and mobile UI/UX design. It's Simplicity and collaboration features.

- **Procreate:** A highly popular app for digital painting and drawing, perfect for creating artwork on the go.

❖ Network devices used by a graphic agency

- **Router:** The router is the central devices that connects the agency's internal network to the Internet. A high- performance router router ensures fast internet speed and reliable. Communication for task like file sharing or client video call.

- **Switch:** A switch connects multiple devices within the office network, allowing them to Communicate with each other. Switch help create wired network to ensure stable and fast connections between Workstation, printers, and servers.

- **Wi-Fi Access Points (APs):** This are devices that allow wireless devices (such as smartphones, laptops, tablets) to connect to a wired network using Wi-Fi. The AP acts as a bridge between the wired network (such as Ethernet) and the wireless clients, providing them with network access and enabling them to communicate over the internet or local networks.

- **Firewall:** A **firewall** is a security system designed to monitor and control incoming and outgoing network traffic based on predetermined security rules.

- **Network Attached Storage (NAS):** This is allowing to multiple devices (Computers, smartphones, tables, etc..) to access and share to data centrally. This specialized file storage device connects to a network.

❖ Best 10 practices to have a better life time for your pc

- **Keep Your Software Update**d: Keep your operating system and software updated regularly. This will help to improve functionality, and confirm better performance.

- **Install Antivirus and Antimalware Programs**: Protect your PC from virus attacks and other malware. The use of antiviruses and antimalware can prevent viruses, spyware, and other rogue programs.

- **Clean Dust Regularly**: Clean the insides of your case and fans after every couple of months to ensure good airflow and cooling.

- **Manage Startup Programs:** It will let your computer get up and running faster and make it operate smoother and quicker**.**

- **Upgrade Your Hardware**: If your PC is slowing down, then upgrade your RAM, shift to SSD, or install an improved graphics card. It can give a whole new life to your computer.

- **Monitor System Performance**: Keep an eye on your PC's CPU, memory, and disk usage to avoid overloading your system. Close unnecessary applications to conserve resources.

- **Clear Cache and Temporary Files:** Temporary files and browser cache, over time, build up and eat up huge storage

- **Uninstall unwanted Programs:** Remove software which you don't find useful or don't need. It's like cleaning out a cluttered drawer — it frees up space and helps your PC run smoother.

- **Backup Your Data:** Regularly save copies of your important files to an external drive or cloud storage. It's a simple step that can save you from the heartbreak of losing valuable photos, documents, or work.

- **Shut Down Properly**: Always power OFF your PC properly. Closed correctly and to prevent potential software or hardware issues.

❖ Best 10 practices to keep your pc safe and secure

- **Backup Your Data Regularly:** This way, if something goes wrong with your computer, your files are safe and easy to recover. Save copies of important files to an external drive or cloud storage.

- **Use Secure Wi-Fi Networks:** Using public Wi-Fi for tasks that require privacy, like online banking. If you must use it, get a VPN to encrypt your connection and protect your data.

- **Limit User Permissions:** Separate accounts for different people using your PC, especially for those who don't need admin access. This way, you reduce the risk of accidental changes or malicious actions on your system.

- **Be Careful with Emails and Links:** Don't open emails or click links from unknown senders. Phishing scams often trick you into sharing personal info, so always be careful and double-check before clicking.

- **Backup Your Data Regularly:** Backing up your data like taking out an insurance policy for your files. That way, if something goes wrong with your computer, you'll still have everything you need safe and sound.

- **Encrypt Sensitive Data:** your sensitive information is like a diary with your personal thoughts. Encrypting it is like locking that diary with a key only you have, so no one can peek inside. Whether it's financial details or personal documents, encryption keeps your data safe from prying eyes.

- **Use Strong, Unique Passwords:** Create strong passwords with a mix of letters, numbers, and symbols, and make sure each one is unique for every account. This way, even if someone guesses one, they won't get into everything else.

- **Enable Two-Factor Authentication (2FA):** This adds an extra step to logging in, so even if someone gets your password, they can't get in without the second verification.

- **Enable a Firewall:** It's like a security guard that helps block unauthorized access and keeps an eye on incoming and outgoing traffic.

- **Install Antivirus Software:** Antivirus software as a security guard for your computer, always on the lookout for potential threats like viruses or malware.

❖ Details about incremental and differential backups

Backing up data is essential for protecting against data loss, ensuring business continuity, and recovering from disasters. Two common types of backups are **incremental** and **differential** backups. Below is an explanation of both, along with their advantages and disadvantages.

Incremental Backup:

Explanation:

An **incremental backup** only backs up the data that has changed since the **last backup**, whether that was a full or incremental backup. For example:

- On Day 1, a full backup is made.
- On Day 2, only the changes since Day 1 are backed up.
- On Day 3, only the changes since Day 2 are backed up, and so on.

Advantages:

- ✓ Efficient Storage Use: Since only changes are backed up, incremental backups take up less storage space compared to full backups.
- ✓ Faster Backup Process: The backup process is faster since only new or modified files are backed up.

Disadvantages:

- ✓ Slower Recovery: To restore data, you need the full backup and all subsequent incremental backups. This can make recovery slower, as multiple backup sets need to be restored.
- ✓ Complexity: Restoring from incremental backups requires more steps and careful tracking of which backups need to be restored in order.

Differential Backup:

Explanation:

A differential backup backs up all the data that has changed since the last full backup. For example:

- On Day 1, a full backup is made.
- On Day 2, all changes since Day 1 are backed up.
- On Day 3, all changes since Day 1 are backed up again, and so on. Unlike incremental, differential backups do not reset each time.

Advantages:

- ✓ Faster Recovery: To restore, you only need the full backup and the most recent differential backup. This makes recovery faster than with incremental backups.
- ✓ Simplicity: Fewer backup sets need to be tracked compared to incremental backups, making restoration easier.

Disadvantages:

- ✓ Increased Storage Use: Since each differential backup includes all changes since the last full back up, they tend to consume more storage space as time passes.

- ✓ Slower Backup Process: The backup process can take longer over time because the number of changes increases, resulting in larger backups.

❖Summary

This coursework covers key computer parts like processors, RAM, storage, and networking devices. It outlines ideal setups for desktops, laptops, and servers used by graphic agencies. It also highlights tips for keeping PCs running smoothly, staying secure, and using effective data backup methods (incremental vs differential).